Causal Analysis Report

BPI_Challenge_2013_open_problems.xes.gz-imf

Decision points:	3858
Choice sets:	4
Transitions fitted:	4
Significant (p<0.05):	0
Generated:	2026-06-23 20:59

Step 1 — Association test (logistic regression + permutation)
H□: choice □ util | choice_set.   reject_H0=True → util predicts routing.

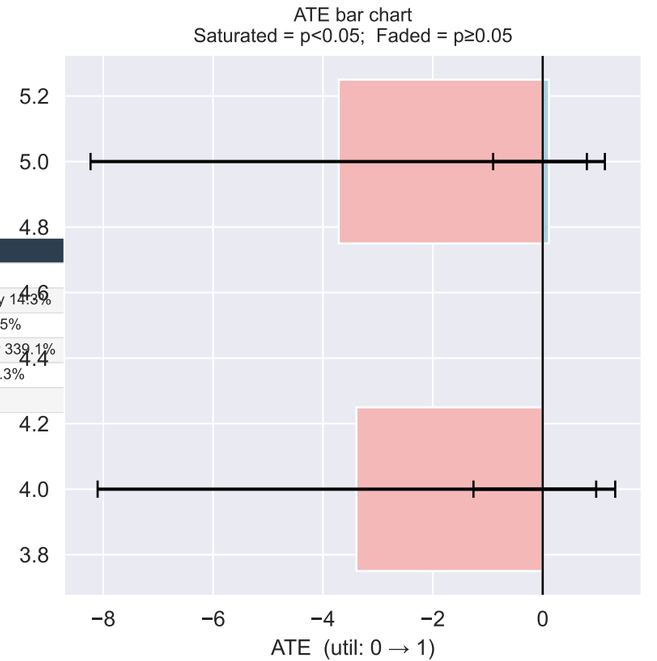
(no data)

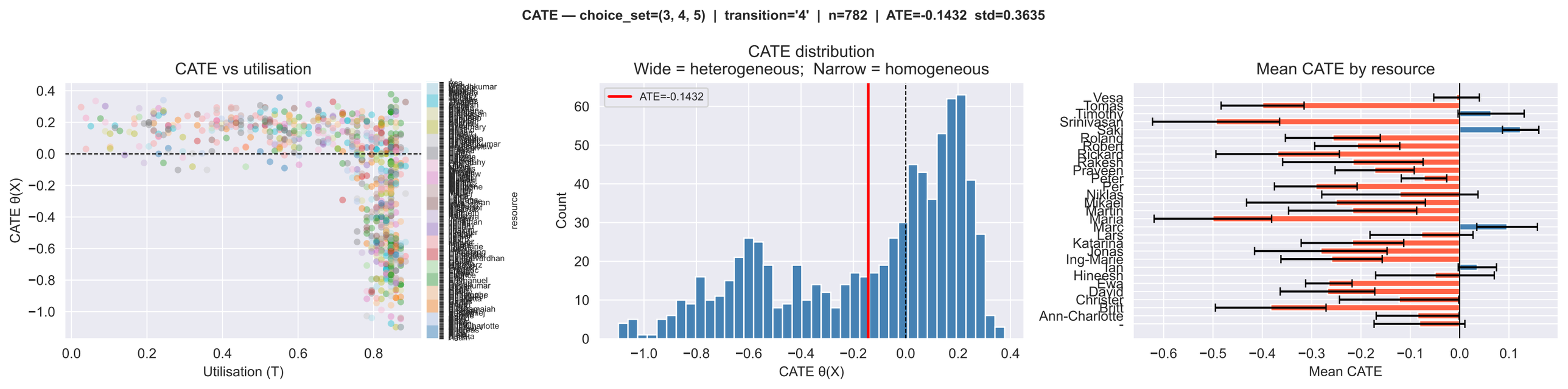
Step 2 — Double ML (partially linear regression, HC3 robust SE)
θ = linear effect of 1-unit ↑ util on P(chose k), after partialling out X.
Equals ATE when the effect is homogeneous. Otherwise: variance-weighted average of CATEs.

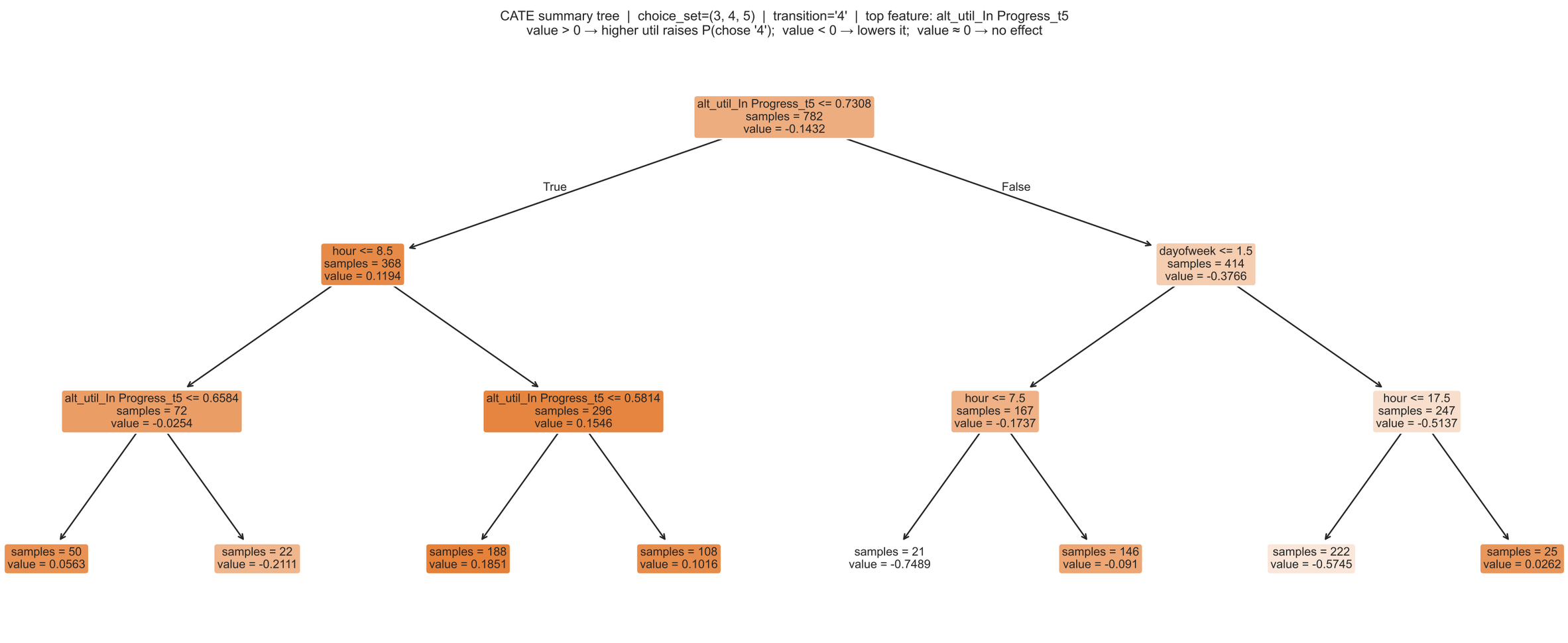
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	theta	se	ci_95	p_value	reject_H0	note
(0, 1)	('τ_t0', 'Assigned_t1')	34	1	—	—	—	nan	nan	—	nan	—	too few observations
(3, 4, 5)	('τ_t3', 'Awaiting Assignment_t4', 'In Progress_t5')	782	4, 5	4	Awaiting Assignment_t4	In Progress_t5	0.122171	0.178321	[-0.2273, 0.4717]	0.4933	False	θ=+0.1222: 1-unit ↑ util ↑ P(t4/Awaiting Assignment_t4) by 12.2%
(3, 4, 5)	('τ_t3', 'Awaiting Assignment_t4', 'In Progress_t5')	782	4, 5	5	In Progress_t5	Awaiting Assignment_t4	0.004863	0.17649	[-0.3411, 0.3508]	0.978	False	θ=+0.0049: 1-unit ↑ util ↑ P(t5/In Progress_t5) by 0.5%
(4, 5)	('Awaiting Assignment_t4', 'In Progress_t5')	719	4, 5	4	Awaiting Assignment_t4	In Progress_t5	-0.355628	0.484924	[-1.3061, 0.5948]	0.4633	False	θ=-0.3556: 1-unit ↑ util ↓ P(t4/Awaiting Assignment_t4) by 35.6%
(4, 5)	('Awaiting Assignment_t4', 'In Progress_t5')	719	4, 5	5	In Progress_t5	Awaiting Assignment_t4	-0.492884	0.511299	[-1.4950, 0.5093]	0.3351	False	θ=-0.4929: 1-unit ↑ util ↓ P(t5/In Progress_t5) by 49.3%
(6, 7)	('τ_t6', 'Closed_t7')	349	7	—	—	—	nan	nan	—	nan	—	too few observations

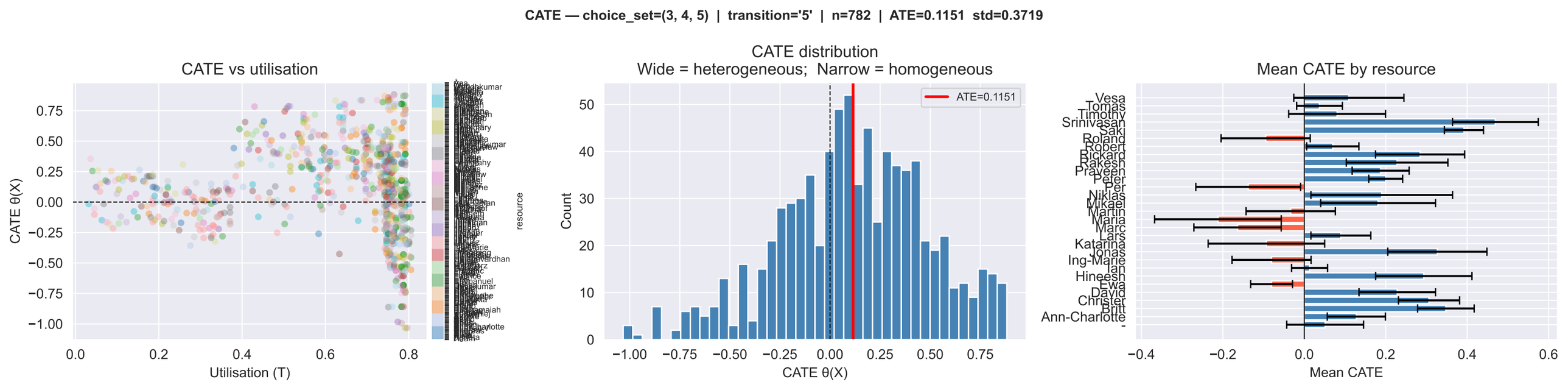
Step 3 — Causal Forest DML (ATE per transition)
ATE = mean of per-sample CATEs. Green rows = H0 rejected (p<0.05).
Less powerful for significance than Double ML, but basis for heterogeneity analysis.

choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	ate	ci_95	p_value	reject_H0	note
(0, 1)	('τ_t0', 'Assigned_t1')	34	1	—	—	—	nan	—	nan	—	too few observations
(3, 4, 5)	('τ_t3', 'Awaiting Assignment_t4', 'In Progress_t5')	782	4, 5	4	Awaiting Assignment_t4	In Progress_t5	-0.143179	[-1.2600, 0.9737]	0.8016	False	ATE=-0.1432: 0→1 util ↓ P(t4/Awaiting Assignment_t4) by 14.3%
(3, 4, 5)	('τ_t3', 'Awaiting Assignment_t4', 'In Progress_t5')	782	4, 5	5	In Progress_t5	Awaiting Assignment_t4	0.115122	[-0.9020, 1.1322]	0.8244	False	ATE=+0.1151: 0→1 util ↑ P(t5/In Progress_t5) by 11.5%
(4, 5)	('Awaiting Assignment_t4', 'In Progress_t5')	719	4, 5	4	Awaiting Assignment_t4	In Progress_t5	-3.39103	[-8.1024, 1.3203]	0.1583	False	ATE=-3.3910: 0→1 util ↓ P(t4/Awaiting Assignment_t4) by 339.1%
(4, 5)	('Awaiting Assignment_t4', 'In Progress_t5')	719	4, 5	5	In Progress_t5	Awaiting Assignment_t4	-3.713344	[-8.2310, 0.8043]	0.1072	False	ATE=-3.7133: 0→1 util ↓ P(t5/In Progress_t5) by 371.3%
(6, 7)	('τ_t6', 'Closed_t7')	349	7	—	—	—	nan	—	nan	—	too few observations




CATE — choice_set=(3, 4, 5) | transition='4' | n=782 | ATE=-0.1432 std=0.3635
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.1432
Count
CATE θ(X)
Mean CATE by resource
Vesa
Tomas
Timothy
Srinivasan
Saki
Roland
Robert
Rickard
Rakesh
Praveen
Peter
Per
Niklas
Mikael
Martin
Maria
Marc
Lars
Katarina
Jonas
Ing-Marie
Ian
Hineesh
Ewa
David
Christer
Britt
Ann-Charlotte
Mean CATE


CATE summary tree | choice_set=(3, 4, 5) | transition='4' | top feature: alt_util_ln Progress_t5
value > 0 → higher util raises P(chose '4'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_ln Progress_t5 <= 0.7308
samples = 782
value = -0.1432
True
False
hour <= 8.5
samples = 368
value = 0.1194
dayofweek <= 1.5
samples = 414
value = -0.3766
alt_util_ln Progress_t5 <= 0.6584
samples = 72
value = -0.0254
alt_util_ln Progress_t5 <= 0.5814
samples = 296
value = 0.1546
hour <= 7.5
samples = 167
value = -0.1737
hour <= 17.5
samples = 247
value = -0.5137
samples = 50
value = 0.0563
samples = 22
value = -0.2111
samples = 188
value = 0.1851
samples = 108
value = 0.1016
samples = 21
value = -0.7489
samples = 146
value = -0.091
samples = 222
value = -0.5745
samples = 25
value = 0.0262


CATE — choice_set=(3, 4, 5) | transition='5' | n=782 | ATE=0.1151 std=0.3719
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=0.1151
Count
CATE θ(X)
Mean CATE by resource
Vesa
Tomas
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Srinivasan
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Rakesh
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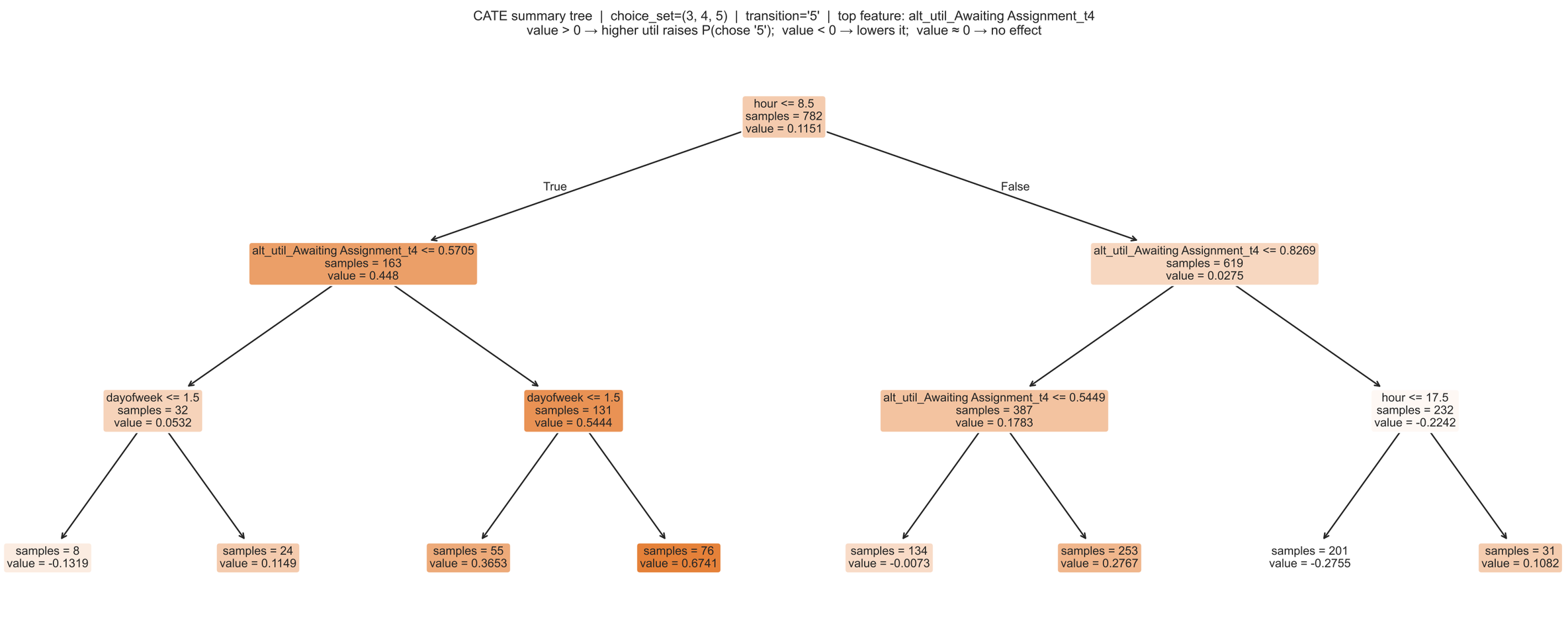
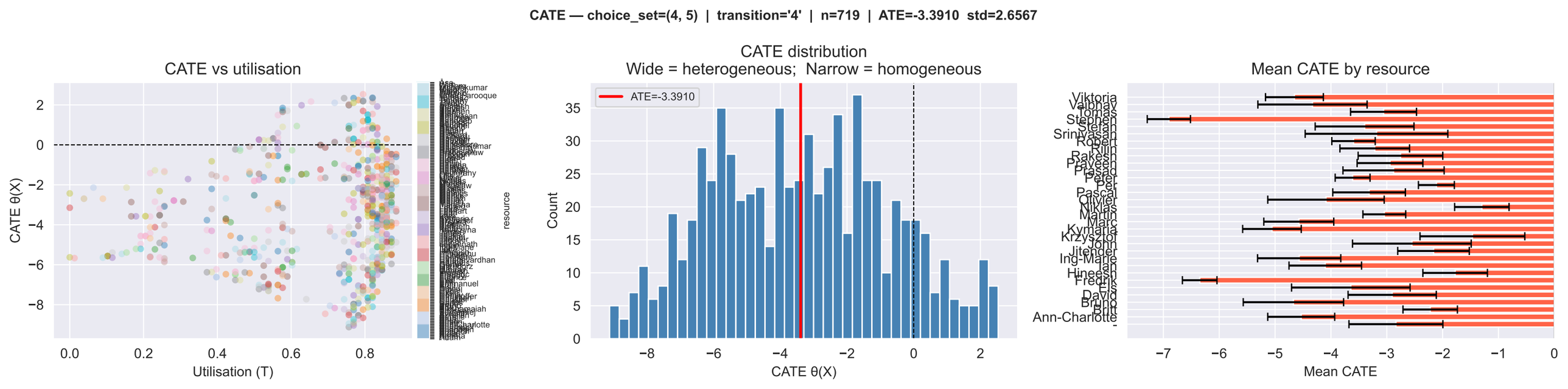
Christer
Britt
Ann-Charlotte
Mean CATE


CATE summary tree | choice_set=(3, 4, 5) | transition='5' | top feature: alt_util_Awaiting Assignment_t4
value > 0 → higher util raises P(chose '5'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 8.5
samples = 782
value = 0.1151
True
False
alt_util_Awaiting Assignment_t4 <= 0.5705
samples = 163
value = 0.448
alt_util_Awaiting Assignment_t4 <= 0.8269
samples = 619
value = 0.0275
dayofweek <= 1.5
samples = 32
value = 0.0532
dayofweek <= 1.5
samples = 131
value = 0.5444
alt_util_Awaiting Assignment_t4 <= 0.5449
samples = 387
value = 0.1783
hour <= 17.5
samples = 232
value = -0.2242
samples = 8
value = -0.1319
samples = 24
value = 0.1149
samples = 55
value = 0.3653
samples = 76
value = 0.6741
samples = 134
value = -0.0073
samples = 253
value = 0.2767
samples = 201
value = -0.2755
samples = 31
value = 0.1082


CATE — choice_set=(4, 5) | transition='4' | n=719 | ATE=-3.3910 std=2.6567
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-3.3910
Count
CATE θ(X)
Mean CATE by resource
Viktoria
Vaibhav
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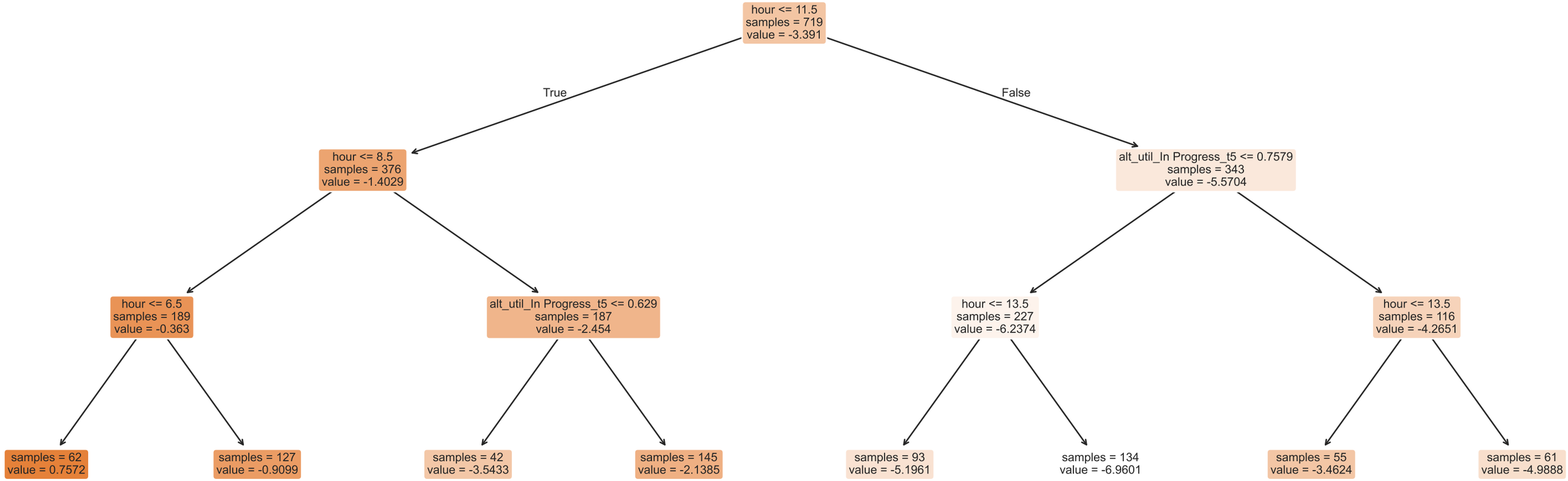
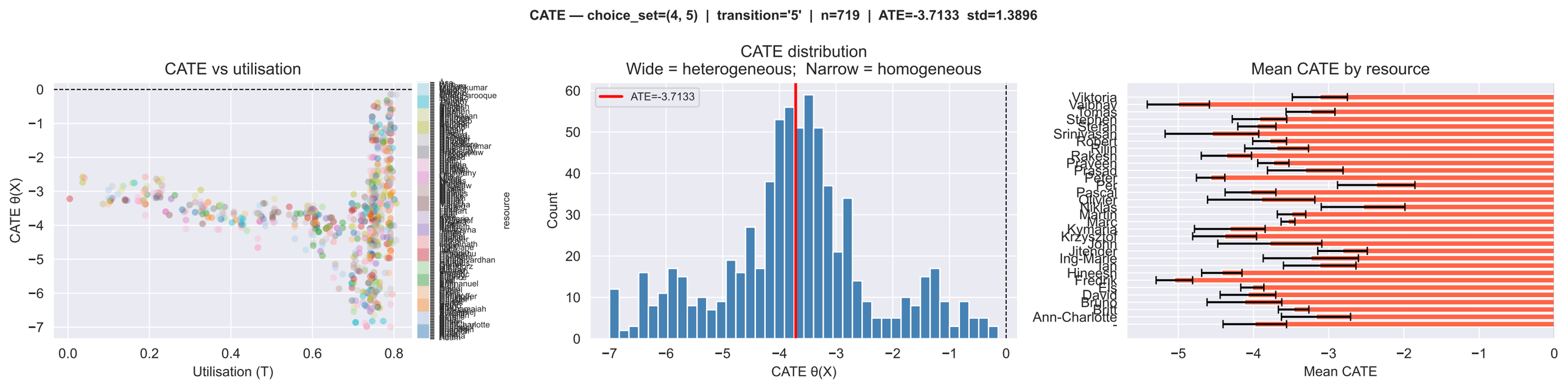
Stephen
Stefan
Srinivasan
Robert
Rilin
Rakesh
Praveen
Prasad
Peter
Per
Pascal
Olivier
Niklas
Martin
Marc
Kymaria
Krzysztof
John
Jitender
Ing-Marie
Ian
Hineesh
Fredrik
Eli
David
Bruno
Britt
Ann-Charlotte
Mean CATE


CATE summary tree | choice_set=(4, 5) | transition='4' | top feature: hour
value > 0 → higher util raises P(chose '4'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 11.5
samples = 719
value = -3.391
True
False
hour <= 8.5
samples = 376
value = -1.4029
alt_util_ln Progress_t5 <= 0.7579
samples = 343
value = -5.5704
hour <= 6.5
samples = 189
value = -0.363
alt_util_ln Progress_t5 <= 0.629
samples = 187
value = -2.454
hour <= 13.5
samples = 227
value = -6.2374
hour <= 13.5
samples = 116
value = -4.2651
samples = 62
value = 0.7572
samples = 127
value = -0.9099
samples = 42
value = -3.5433
samples = 145
value = -2.1385
samples = 93
value = -5.1961
samples = 134
value = -6.9601
samples = 55
value = -3.4624
samples = 61
value = -4.9888


CATE — choice_set=(4, 5) | transition='5' | n=719 | ATE=-3.7133 std=1.3896
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-3.7133
Count
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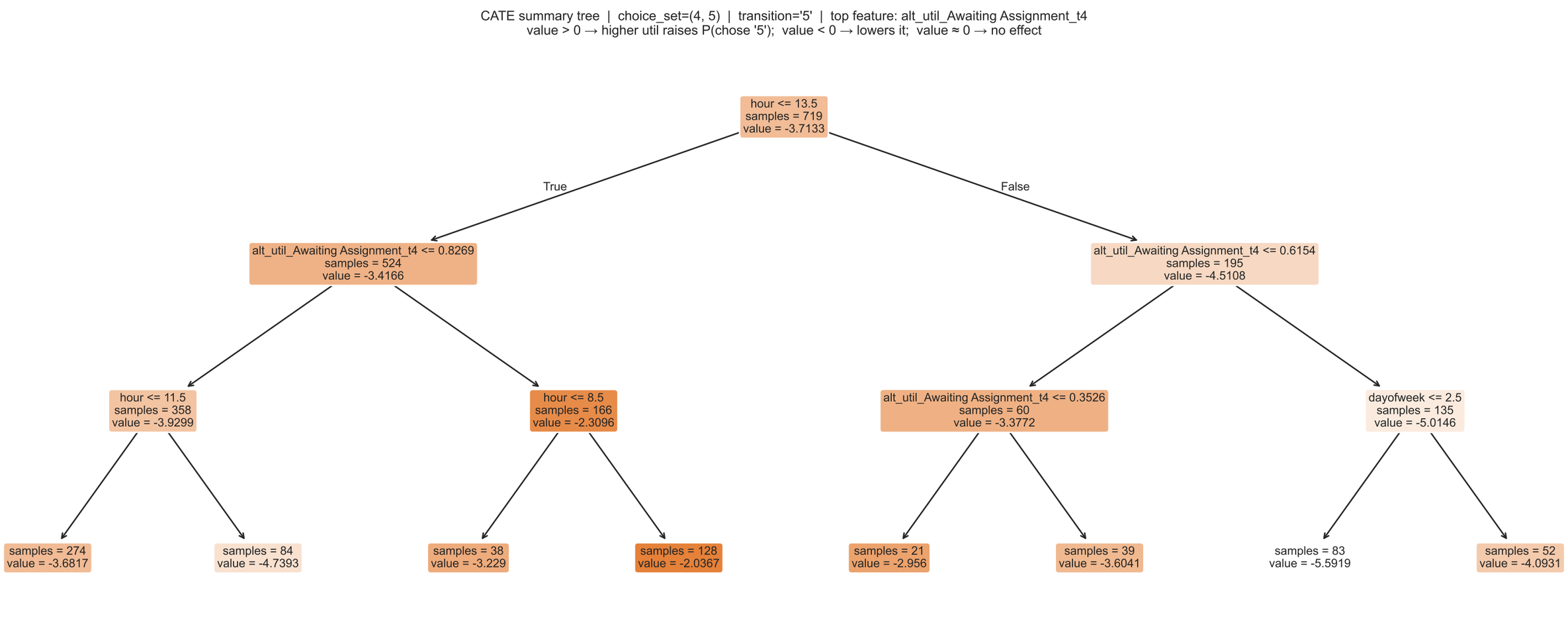
Mean CATE


CATE summary tree | choice_set=(4, 5) | transition='5' | top feature: alt_util_Awaiting Assignment_t4
value > 0 → higher util raises P(chose '5'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 13.5
samples = 719
value = -3.7133
True
False
alt_util_Awaiting Assignment_t4 <= 0.8269
samples = 524
value = -3.4166
alt_util_Awaiting Assignment_t4 <= 0.6154
samples = 195
value = -4.5108
hour <= 11.5
samples = 358
value = -3.9299
hour <= 8.5
samples = 166
value = -2.3096
alt_util_Awaiting Assignment_t4 <= 0.3526
samples = 60
value = -3.3772
dayofweek <= 2.5
samples = 135
value = -5.0146
samples = 274
value = -3.6817
samples = 84
value = -4.7393
samples = 38
value = -3.229
samples = 128
value = -2.0367
samples = 21
value = -2.956
samples = 39
value = -3.6041
samples = 83
value = -5.5919
samples = 52
value = -4.0931

CATE statistics summary
cate_std = heterogeneity; pct_positive/negative = share of cases where util helps/hurts; top_feature = main moderator

choice_set	transition	n	ate	cate_std	cate_min	cate_p25	cate_p50	cate_p75	cate_max	pct_positive	pct_negative	top_feature
(3, 4, 5)	4	782	-0.143179	0.36351	-1.097957	-0.455099	0.0007	0.163392	0.377987	50.3	49.7	alt_util_In Progress_t5
(3, 4, 5)	5	782	0.115122	0.371873	-1.032847	-0.124242	0.113743	0.390486	0.88275	65.1	34.9	alt_util_Awaiting Assignment_t4
(4, 5)	4	719	-3.39103	2.656661	-9.121245	-5.587393	-3.469497	-1.470801	2.53315	11.0	89.0	hour
(4, 5)	5	719	-3.713344	1.389612	-6.997356	-4.425487	-3.68827	-3.081832	-0.140612	0.0	100.0	alt_util_Awaiting Assignment_t4

Key Assumptions & Interpretation Notes

Backdoor criterion (all methods)

X = {hour, dayofweek, n_alternatives, res_*} must block all confounding paths T ← C → Y.

Any variable that affects both util and routing must be in X. Unobserved confounders

cannot be detected and will bias all estimates.

SUTVA (all methods)

One case's routing does not affect another's treatment or outcome.

Violated if resource occupation by one case changes the util seen by a concurrent case.

Positivity (all methods)

Every covariate combination X must have positive probability for all util levels T.

Diagnosed by the covariate balance (SMD) in the backdoor check.

Linearity (Double ML only)

θ is the best-fit linear slope. If the effect varies with X, θ is a variance-weighted

average of CATEs — valid as a summary, but not the simple mean.

Double ML is more powerful than the forest for testing 'is there any effect?'

Honest splitting (Causal Forest only)

Tree structure and leaf estimates use separate subsamples. Yields valid CIs but

halves effective sample size → forest CIs are wider than Double ML's.

A significant Double ML θ with an insignificant forest ATE is not a contradiction.

What the analysis cannot establish

Causal direction (assumes util → choice, not the reverse).

External validity beyond the observed covariate distribution.

Dynamic/lagged effects across cases over time.

Full documentation: ASSUMPTIONS.md